

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

Unit – 4 : Session – 6 : SLO – 1 | Hardwired Control — Choose the Best

■ I. Multiple Choice

1. b) Implemented with logic gates — a hardwired control unit is pure digital logic (gates, decoders, flip-flops).
2. b) Decoder — the instruction decoder interprets the opcode and drives signal generation.
3. b) Simple, fixed instruction sets — hardwired logic is economical when the ISA is small and stable (RISC-style).
4. c) Faster operation — signals come straight from logic gates with no control-memory read, so hardwired beats microprogrammed on speed.
5. c) Control memory — control memory belongs to microprogrammed control, not hardwired.
6. b) Current instruction and processor state — signals are functions of the opcode, step counter, and condition flags.
7. c) RISC processors — their simple, fixed instruction sets make hardwired control practical; modern GPUs similarly use hardwired logic for speed.
8. b) Requires hardware modification — the logic is physically wired, so adding an instruction means redesigning the circuit.
9. b) Hardwired control — the control unit is designed as a finite state machine stepping through instruction states.
10. b) Propagation delay of logic gates — the delay through the combinational logic sets the minimum cycle time.